YEAR 10 SCIENCE
CHEMICAL SCIENCE
TEST 1

NAME: SOLUTIONS

MARK: $\frac{\quad}{38}$

The atomic structure and properties of elements are used to organise them in the Periodic Table.

Mark	ND	NW	C	O
Mark Range	0-7	8-16	17-27	28-38

QUESTION	ANSWER
1	B
2	D
3	C
4	A
5	C
6	B
7	B
8	A
9	A

1. The ***atomic number*** of an element represents:

   (a) the number of electrons.
   (b) the number of protons.
   (c) the number of neutrons.
   (d) the number of protons added to number of electrons.

2. The nucleus of an atom consists of:

   (a) electrons and protons.
   (b) electrons and neutrons.
   (c) protons only.
   (d) protons and neutrons.

3. How many ***protons*** does a sodium atom have?

   (a) 12
   (b) 10
   (c) 11
   (d) 5

4. The ***atomic number*** of phosphorus is:

   (a) 15
   (b) 16
   (c) 31
   (d) 30

5. Which of the following are all ***metals***?

   (a) Calcium, strontium and silicon.
   (b) Fluorine, sodium and argon.
   (c) Iron, potassium and aluminium.
   (d) Boron, lithium and beryllium.

6. Which of the following is a ***noble or inert gas***?

   a. oxygen
   b. helium
   c. hydrogen
   d. carbon dioxide

7. Which of the following are all ***transition metals***?

   (a) Li, Mn, Ca.
   (b) Mn, Fe, Cu.
   (c) F, Cl, Br.
   (d) Na, K, Fe.

8. Which of the following are all ***alkali metals***?

(a) Li, Na, K.
(b) Mg, Ca, Na.
(c) He, Ne, Ar.
(d) Cu, F, Mn.

9. The charge of an electron is:

(a) negative.
(b) neutral.
(c) positive.
(d) nothing.

Short Answer **Write the answer to the questions in the spaces provided.**

1. In the table below, choose the correct meaning for each word given. ***Write the letter of the correct meaning next to the word.***

WORD	CORRECT LETTER	MEANINGS
atom	E	A - two or more atoms chemically combined.
molecule	A	B - particle with no charge.
compound	D	C - a row of the periodic table.
neutron	B	D - two or more elements chemically combined.
period	C	E - building block of matter.
group	F	F - column of the periodic table.

(6)

2. Classify the following elements as ***alkali metals, transition metals, metalloid (semi-metals), or halogens*** or ***noble (inert) gases***.

He Si Na Fe Cl K Br Ar Li Cu Ge I

Alkali metal	Transition Metal	Metalloid (Semi-Metal)	Halogen	Noble (Inert) Gas
Na	Fe	Si	Cl	He
K	Cu	Ge	Br	Ar
Li			I	

(6)

($\frac{1}{2}$ mark each)

3. Write the symbols or names for the following elements.

ELEMENT	SYMBOL
aluminium	Al
sulphur	S
nitrogen	N
potassium	K
copper	Cu
Boron	B
Silicon	Si
Chlorine	Cl
Gold	Au
Magnesim	Mg

(5)

4. Complete the following table.

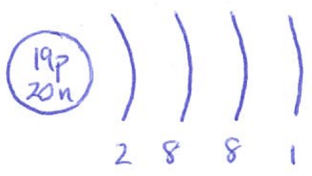
ELEMENT	NUMBER OF PROTONS	NUMBER OF NEUTRONS	NUMBER OF ELECTRONS
$^{31}_{15}P$	15	16	15
$^{14}_{7}N$	7	7	7
$^{23}_{11}Na$	11	12	11
$^{19}_{9}F$	9	10	9

(6)

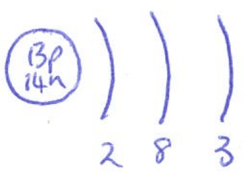
5. Draw diagrams to show the electron configurations of the following elements. Remember to include the particles in the nucleus.

(a) $^{16}_{8}O$

(b) $^{39}_{19}K$



(c) $^{27}_{13}Al$



Nucleus - 1 mark

Shells - 1 mark.

(6)